

Task-Directed Releases Beside a Fixed Prediction Service:

Measuring Additional Sensitive Recovery for Single and Colluding Recipients

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Abstract—An organisation already publishes a prediction service whose outputs downstream recipients consume and cannot be asked to re-accept. We study what it can release *beside* those fixed outputs for one recipient: a channel that must serve a permitted task while limiting what that recipient, and that recipient colluding with another, can newly infer about sex and race. We formalise this purpose-specific release problem, give it an implemented interface and measurement contract—per-recipient and coalition views, independently fitted attackers, additional disclosure always reported against the fixed service baseline, and negative increments retained unclipped—and use it to evaluate a task-directed finite release optimised under explicit conditional-disclosure constraints. On repeatedly used development data, the selected release improves permitted-task loss over the strongest channel this project had by 0.0163 nats with paired household bounds excluding zero, and all eight sensitive point estimates favour it; but seven of the eight adjusted non-inferiority bounds do not resolve, so the registered competitiveness conjunction is not established, and a matched deterministic release without the stochastic machinery performs comparably. We report what the comparison does not isolate: the task objective used authorised supervision the comparator never received, and neither randomisation nor coalition conditioning earned its registered attribution. We also correct published claims of our own, including a linear-erasure guarantee that a counterexample refutes, and state precisely which guarantees are finite-model rather than population statements.

I. INTRODUCTION

A bank collects mortgage applications and uses them for several purposes. An underwriter must not see applicant race; a fair-lending auditor must. A pricing analyst sits somewhere in between. Each purpose admits a different set of permissible attributes, and this is the dilemma of *purpose limitation* codified in data-protection regulation and, at the representation level, of contextual integrity.

Most of the machine-learning literature attacks this problem by owning the representation: it edits, projects or retrains until an attribute is hard to recover. In deployment the harder version is often the one where *the organisation has already published something*. A prediction service is live, its outputs are inside the recipients' systems, and they cannot be recalled or renegotiated. The only decision left is what to place *beside* the fixed outputs, for whom, and how to measure what that addition gives away.

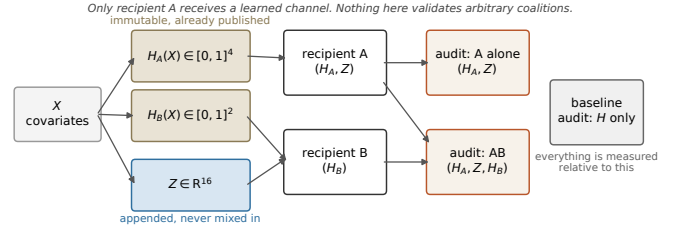


Fig. 1. The release interface. H_A and H_B are already published and are never modified. A channel is appended to recipient A's wire alone. Three views are audited: A alone, the coalition AB, and the H-only baseline against which every increment is measured.

This paper formalises and evaluates that release problem. A service publishes probability vectors H_A to one recipient and H_B to another; a mechanism may add a channel for the first recipient only. The permitted task is a residence-mobility prediction that recipient is authorised to perform; the protected targets are sex and race, forbidden to both recipients individually and to the two acting together.

Contributions:

- **The release problem and its measurement contract (§II):** per-recipient and coalition audit roles, independently fitted attackers, additional disclosure always reported against the fixed service baseline, negative increments retained unclipped, per-endpoint results never averaged.
- **An auditing result that survives checking (§III):** a prevalence-weighted aggregate one-hot regression score can conceal a much larger sensitive direction, which a class-aware dominant-axis audit exposes; we give the exact convex-combination identity behind it, and the precise limits of the linear guarantees it is often paired with.
- **A task-directed finite release (§V)** optimised under explicit conditional-disclosure constraints, with its optimisation assumptions, its relationship to established convex programmes, and an exact fixture separating what it can and cannot do.
- **A controlled evaluation (§VI)** in which the selected release improves the permitted task over the strongest prior channel with bounds excluding zero, does *not* establish the

registered competitiveness conjunction, and is matched by a simpler deterministic baseline—together with the supervision asymmetry that comparison does not isolate.

- **Corrections to our own published claims (§C)**, including a universal regression-to-classification guarantee that an exact counterexample refutes.

II. THE RELEASE PROBLEM AND THE MEASUREMENT CONTRACT

a) Views: Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published first: $H_A(X) \in [0, 1]^4$ and $H_B(X) \in [0, 1]^2$. A mechanism may add a channel Z for A only, so A holds (H_A, Z) , B holds H_B , and the coalition holds (H_A, H_B, Z) . The policy authorises A for the residence task and forbids sex and race to A , to B and to the coalition.

b) Threat model: The adversary holds labelled data from the release year, fits attack families on its own pool, selects on a validation pool, and sees only its legal view. It knows the mechanism, the codebook and the channel law; what it does not hold is the person-specific draw. Attacks are fitted on the *unrounded continuous* service predictions, never on a coarsened view, because any conditioning partition used inside an optimiser is a property of that optimiser and not of the released interface.

c) Single-token semantics: The released object is one sampled token per person, drawn once and reused. Publishing the channel row itself, or redrawing per query, is a different mechanism with different disclosure, and we never score one as if it were the other. Expected scoring averages *losses* over the token distribution, not predicted probabilities; the two differ, and only the first is the expectation of what a recipient experiences.

d) What a number means here: Every recovery figure is the log-loss reduction an independently fitted, validation-selected attacker achieves relative to the same family restricted to H . It is a floor on what is recoverable, never a ceiling, and it is not an estimate of conditional mutual information [1]: a restricted predictor can improve after receiving a deterministic feature $Z = f(H)$ even where $I(S; Z | H) = 0$, and a failed attack bounds nothing from above.

III. WHAT LINEAR AUDITS DO AND DO NOT CERTIFY

The evaluation contract above is a response to a concrete measurement failure we documented in this line of work, and the failure is worth stating because it motivates every audit choice that follows.

a) Aggregate scores hide rare-class leakage: A common compliance reading is an aggregate one-hot regression score $R^2(h, A)$ against a multi-class protected attribute. For an imbalanced attribute this aggregate is a prevalence-weighted convex combination of the per-class one-versus-rest scores,

$$R^2_{\text{onehot}} = \sum_k w_k R^2_{\text{OvR}, k}, \quad w_k = \frac{\pi_k(1-\pi_k)}{\sum_j \pi_j(1-\pi_j)},$$

TABLE I
AGGREGATE ONE-HOT VERSUS DOMINANT-AXIS AUDITING ON THE MULTI-CLASS PROTECTED ATTRIBUTE, MEAN OVER PAIR-SEEDS OF THE CANONICAL CHECKPOINTS. THE AGGREGATE UNDERSTATES THE RECOVERABLE DIRECTION, MOST ON THE IMBALANCED ATTRIBUTE. HISTORICAL EVIDENCE FROM THE ENCODER LINE DESCRIBED IN §VIII.

dataset	pairs	mean R^2_{onehot}	mean R^2_{DA}	gap
Adult	12	0.0080	0.0114	+0.0034
HMDA	9	0.0192	0.0609	+0.0417
Diabetes	12	0.0125	0.0179	+0.0054

so a rare class contributes almost nothing to the number a compliance report prints. Auditing the *dominant axis* instead—the worst single direction across classes—exposes what the aggregate smooths away. On the imbalanced mortgage attribute the gap is large (Table I): mean $R^2_{\text{DA}} = 0.0609$ against $R^2_{\text{onehot}} = 0.0192$ over nine multi-class pair-seeds, and in one configuration the aggregate reads 0.027 where the dominant axis reads 0.288. The identity itself was validated on every multi-class pair-seed, maximum absolute residual 0.0021.

b) Three limits we now state explicitly: First, a **zero linear score does not bound classification**. We previously asserted that zero affine least-squares R^2 forces every linear threshold classifier to majority accuracy. That is false, and a twenty-row counterexample settles it: both conditional feature means are zero, so $\text{Cov}(h, A) = 0$ and the optimal affine predictor is constant with $R^2 = 0$, yet thresholding at $h > 0$ classifies 18/20 correctly against a 50% majority baseline. The claim is withdrawn and the corresponding API now refuses to run. The valid statement is narrower and about squared loss only: zero cross-covariance means affine least-squares prediction cannot improve on the optimal constant predictor on that distribution.

Second, **composition is exact only at exact zero**. Two views with exactly zero cross-covariance with the same signal retain exactly zero under concatenation. Approximate leakage does not inherit this: $h_1 = N + \delta S$ and $h_2 = N - \delta S$ each carry $R^2 = \delta^2/(1 + \delta^2)$ and their difference recovers S . Neither statement supplies independence or any guarantee against a nonlinear attack.

Third, **more retained rank is not more task performance, and fewer flagged attacks is not low absolute disclosure**. We report absolute recovery beside every increment for this reason.

IV. RELATED WORK

a) Erasure the releaser owns: Linear concept erasure gives an exact guarantee against linear adversaries on a representation the releaser may edit [2], and task-relevant variants preserve authorised covariance [3]. Adversarial fair and transferable representations are due to Madras et al. [4]; adversarial removal is known to leave recoverable information [5], and least-privilege learning has fundamental limits [6]. The closed-form line of Sadeghi et al. [7]–[9] and the conditional extension U-FaTE [10] solve the same shape of problem when the representation is the releaser’s to change. All of these

assume what our setting denies: here the published output is immutable and the only decision is what to append.

b) Finite-alphabet privacy–utility optimisation: The convex programme of §V is established. Rassouli and Gündüz [11] give the perfect-privacy characterisation, the nullspace feasibility criterion and the linear-programme form of the zero-budget linear-cost problem; the privacy funnel of Makhdoumi et al. [12] is nonconvex precisely because of a utility constraint we do not impose. Multi-user and side-information formulations of utility–privacy tradeoffs [13], [14] design a sanitised release for many consumers. We claim no new optimisation result: the nullspace criterion and the convexity facts are theirs, and what is ours is the release contract, the per-recipient and coalition constraint structure, and an evaluated operating point. Related conditional-independence and information-bound machinery we rely on for scoping comes from the kernel conditional-independence literature [15], [16], cross-fitted nuisance estimation [17], and the formal limits of mutual-information estimation [1]. The data are Folktables-style ACS microdata [18].

V. A TASK-DIRECTED FINITE RELEASE

a) Construction: A label-free code $T = g(X_A)$ over permitted A -side inputs is mapped to a finite output alphabet by a row-stochastic channel Q , and the recipient receives one sampled token. Writing $C_A = H_A$ and $C_{AB} = (H_A, H_B)$ for the conditioning views, the programme is

$$Q \geq 0, Q \mathbf{1} = \mathbf{1} \quad \sum_{t,z} p(t) Q_{tz} D_{tz} \quad \text{s.t.} \quad I_Q(S_r; Z | C_r) \leq \delta_r,$$

with a cost table D fixed before optimisation. Under the mechanism’s Markov structure $Z - T - (S, C)$ and a fixed fitted law, each conditional mutual information is affine in Q inside a jointly convex relative entropy, so the feasible set is convex and the objective linear.

b) Attribution, honestly: This is an *established* convex programme applied to a new release contract, not a new optimisation result. The finite-alphabet privacy–utility problem, the nullspace feasibility criterion and the linear-programme form at zero budget are prior work; the privacy funnel is nonconvex precisely because of a utility constraint we do not impose. Optimising the decoder jointly with Q , or maximising a task mutual information, destroys the convexity, and we do neither. What is ours is the release contract, the per-recipient and coalition constraint structure, and the evaluated operating point.

c) What the constraint is, and is not: δ_r binds a conditional mutual information in a *fitted finite model* whose conditioning view is a coarsened partition. It is not a population guarantee and not a statement about the continuous service vectors: in a validated fixture, binning a conditioning view drops a full $\log 2$ of measured leakage to exactly zero. All audited attacks therefore use continuous H , and the budgets are never quoted as measured increments over the prior channel.

TABLE II

THE SELECTED RELEASE Q AGAINST J : PAIRED HOUSEHOLD BOOTSTRAP, SIMULTANEOUS ADJUSTED BOUNDS OVER THE FROZEN FAMILY OF 706. THE TASK IMPROVEMENT RESOLVES; SEVEN OF THE EIGHT SENSITIVE ENDPOINTS DO NOT. DEVELOPMENT DATA.

endpoint	wt.	estimate	lower	upper	resolves
<i>permitted task: residence log loss versus J (negative = better)</i>					
residence unw.		−0.01625	−0.02728	−0.00522	yes
residence pers.		−0.01639	−0.02917	−0.00362	yes
<i>additional sensitive recovery versus J (negative = less)</i>					
A/sex unw.		−0.00961	−0.02159	+0.00236	no
A/race unw.		−0.00567	−0.01717	+0.00583	no
AB/sex unw.		−0.01103	−0.02199	−0.00007	yes
AB/race unw.		−0.00394	−0.01552	+0.00765	no
A/sex pers.		−0.00246	−0.01628	+0.01135	no
A/race pers.		−0.00482	−0.01760	+0.00797	no
AB/sex pers.		−0.00620	−0.01887	+0.00647	no
AB/race pers.		−0.00463	−0.01694	+0.00768	no

d) Why a replacement can beat an extension: An exact fixture bounds what any “does the new channel add anything?” test can conclude. With H constant, a prior channel $J = (Y, S)$ carrying both the useful bit and the sensitive bit, and a candidate $T = Y$: appending T to J adds exactly zero task information, while *replacing* J by T preserves the task in full and removes the sensitive bit entirely ($\log 2$ nats). A test that measures the first can return zero while the second is strictly better. This is a design example, not an ACS finding.

VI. EVALUATION

a) Setup: California ACS one-year public-use microdata, ages 19–34, household-separated pools for representation fitting, task fitting and validation, attacker fitting and validation, and evaluation. Uncertainty is a paired household bootstrap over 5,473 households, aggregated across three anchors that share people, with simultaneous Bonferroni-adjusted bounds over a frozen family of 706 contrasts ($z = 3.97$). All numbers in this section are **development** evidence on repeatedly used pools and are conditional on the fitted releases.

b) The selected release improves the permitted task: The utility-route nominee Q reduces residence log loss relative to the strongest channel this project had, J , by 0.01625 nats unweighted and 0.01639 person-weighted, with adjusted bounds excluding zero in both weightings (Table II, Figure 2).

c) The competitiveness conjunction is not established: All eight sensitive point estimates favour Q over J , but only one adjusted non-inferiority bound resolves (AB/sex unweighted, upper -0.00007). The registered conjunction therefore fails. **This is not evidence of excess disclosure:** seven endpoints are unresolved at this precision, and an unresolved interval is not a finding in either direction.

d) A simpler baseline matches it: D_{17} uses the same code and the same 17-action dictionary with no privacy constraint and no stochastic optimisation. It attains a slightly *better* task point estimate than Q and *lower* sensitive point estimates in seven of eight cells (Table III). All ten corresponding adjusted differences are unresolved, so we do not claim D_{17} is better, and we equally do not claim the two are equivalent. D_{33} , with a larger dictionary, has better task point estimates and mixed

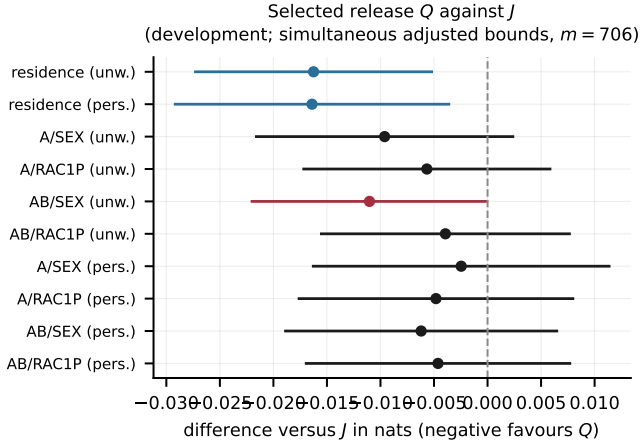


Fig. 2. The same comparison. Every sensitive point estimate favours Q , and seven of eight intervals cross zero: *unresolved*, which is neither demonstrated protection nor demonstrated harm.

TABLE III

RELEASES COMPARED AGAINST J ON THE PERMITTED TASK AND ON ADDITIONAL SENSITIVE RECOVERY (UNWEIGHTED POINT ESTIMATES; POSITIVE TASK VALUES FAVOUR THE RELEASE). H CARRIES NO CHANNEL, SO IT IS WORSE AT THE TASK AND LEAKS LESS. DEVELOPMENT DATA; POINT ESTIMATES ONLY.

release	task vs J	worst sens. vs J
H only (no channel)	-0.02570	-0.00550
J (prior channel)	-0.00000	-0.00000
Q : task-directed, constrained	+0.01625	-0.00246
D_{17} : deterministic, same dictionary	+0.01720	-0.00285
D_{33} : deterministic, larger dictionary	+0.01794	-0.00364
continuous task prediction	+0.01962	+0.00123

sensitive comparisons; its cap feasibility was not established and it is not a dominating method. The honest reading is that on this evidence **the stochastic machinery has not been shown to buy anything over a matched deterministic release.**

e) Baselines, and what each one was allowed to see:

The comparison set is deliberately wider than J . *Deterministic action releases* (D_{17} , D_{33}) use the same label-free code and action dictionary with no privacy constraint. A *direct code* release publishes the code index itself. A *continuous task prediction* publishes the real-valued authorised prediction and is the strongest task point in the table, with the worst sensitive comparison of the group. *Supervised erasure adaptations* (LEACE and SPLINCE) were fitted with protected labels on the same standardised inputs plus the clipped frozen residence-teacher logit; SPLINCE additionally preserves residence, income and employment covariance, so it sees authorised task labels that J never saw. Two fitting pools were prepared in advance—one matching the mechanism’s own fresh-label rows and one the broader union—so that no post-outcome choice between fairness interpretations was needed, with each method’s own complete-case mask disclosed rather than assumed identical. These adaptations completed, but neither supplied an *eligible* nominee for the utility route, which is why

the stricter all-family diagnostic is reported as blocked rather than as passed or failed. An unavailable comparator family is not a verdict about that family.

f) What the comparison does not isolate: Q ’s objective used authorised residence supervision. J never received that training objective. The comparison therefore establishes neither unseen-task transfer nor an isolated benefit of the new optimiser, and a supervised task result must not be described as an untouched-task result. The continuous task predictor and the supervised eraser adaptations are reported with their actual label access and pools for the same reason.

g) Neither refinement earned its claim: Neither risk refinement nor coalition conditioning obtained its registered attribution: the nominated release is *local*. Beating J alone cannot attribute a benefit to the coalition machinery.

h) Why an earlier explanation was wrong: A predecessor experiment found that a label-free code of the permitted inputs added nothing measurable once the published predictions were in the recipient’s view, and read this as the published output already carrying the available signal. A subsequent controlled diagnosis separated the two candidate explanations and found the opposite: the *unquantized* input view attains strictly lower residence loss than J , with adjusted bounds below zero, while a 64-state code of the same inputs gives that advantage away. The obstacle was quantization, not redundancy, and the earlier universal reading is withdrawn (§C).

VII. LIMITATIONS, AND A DIRECT ANSWER

Development, not confirmation. The 2018 pools have been used repeatedly across this line of work. Three anchors and simultaneous intervals do not restore independence from prior research, and no result here is a confirmation.

Unresolved is not equivalence, and not harm. Seven of eight sensitive comparisons are unresolved. We do not convert them into either claim.

Fitted-model budgets are not population guarantees, coarse conditioning is not conditioning on the continuous service vectors, and negative finite-attacker increments do not demonstrate negative conditional mutual information.

Attacks bound from below only. Three searches for a stronger attacker across this project each failed to beat the standard slate, which bounds what those searches found, not what exists.

Audits do not transport across hardware below about 0.003 nats, so comparisons use same-host references.

a) What was established: A task-directed release beside a fixed service improved the permitted task over the strongest channel we had, with bounds excluding zero, while every sensitive point estimate moved in the favourable direction and seven of eight remained unresolved. **What was not established:** the registered competitiveness conjunction, any advantage of the stochastic machinery over a matched deterministic release, any coalition-specific benefit, and any unseen-task claim.

VIII. RELATION TO THE EARLIER ENCODER LINE

This work descends from a purpose-conditioned representation line in which one shared frozen backbone carried per-

purpose adapters, each warm-started from a closed-form erasure projection [2] and trained under a per-purpose linear leakage bound. That setting differs from this one in its release contract: there the releaser owned and re-emitted the representation, here the service output is immutable and the decision is what to append. The auditing result of §III and the composition limits come from that line and survive checking; its training results are not re-used as evidence for the present mechanism, because they tested a different algorithm against a different claim. Appendix A gives the correspondence.

OPEN SCIENCE

Every number in this paper is generated from committed evidence at pinned commits, and each printed digit traces to a (commit, path, SHA-256) triple recorded in the artifact. The artifact contains the table and figure generators, the independent verification scripts, the claim ledger, the correction log, synthetic fixtures, schemas, aggregate result tables and provenance hashes, together with a recipe for obtaining the public ACS microdata this work uses. No restricted data is redistributed and no individual-level records are included. Anonymised artifacts will be shared within the window the call specifies.

LLM USAGE CONSIDERATIONS

This work was substantially assisted by a large language model, and describing that as editorial help would be inaccurate. A model was used to propose and refine experimental designs, to write and refactor the experiment and analysis code, to draft and revise this manuscript, and to audit the studies’ own reports against their source code and stored outputs. Registrations, margins and stopping rules were committed before outcomes. Model-generated analyses were checked by independently written scripts that recompute each reported quantity from the smallest machine-readable source, and several statements in this paper exist because that recomputation contradicted an earlier write-up—the withdrawn explanation in §VI and the retired guarantee in §III are two. The authors take full responsibility for every claim, number and omission in this work, including those first drafted by a model; nothing was accepted on the model’s assertion alone, and every load-bearing figure is reproducible from the artifact.

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APPENDIX A

LINEAGE AND WHAT EACH STUDY TESTED

The encoder line and the present release line share an evaluation philosophy and a protected-attribute audit, not an algorithm. Claims from the encoder line used here are limited to the dominant-axis audit, the convex-combination identity, the retired guarantee and the composition boundary.

The encoder line’s training headline totals are same-grid post hoc development evidence: the dual floor and warm-up rule were developed after examining an earlier evaluation on the same grid, and neither was validated on a held-out subset of purpose/attribute/seed triples. A later prospective registration does not retrospectively change that scope.

APPENDIX B COUNTS AND ACCOUNTING

The task-directed programme’s primary matrix is 81 nominal maps (54 positive, 18 zero, 9 unconstrained); an independent replay covered all 81. Of 162 nominal maps in the full matrix, 126 were scheduled and completed and 36 were prospectively unscheduled under the registered resource schedule; the completed maps reduce to 125 exact kernel groups. There are 297 current audits and evaluations. These are resource and accounting figures, not scientific contributions.

APPENDIX C CORRECTIONS THIS PAPER MAKES

- 1) **The universal regression-to-classification guarantee is withdrawn.** Zero affine R^2 can coexist with 90% threshold accuracy; the valid statement is the narrower squared-loss one (§III).
- 2) **A shared union eraser does not demonstrate retention of conflicting purposes.** In the recorded comparison the shared union destroys both tasks ($R^2 \approx 0$ on each), while separate per-purpose erasure retains both (0.9996, 0.9997); the restricted arm ties that comparator rather than beating it.
- 3) **“The published output already carries the signal”** is withdrawn: a controlled diagnosis attributes the effect to quantization, not redundancy (§VI).
- 4) **A proposed near-optimality certificate is not an implemented result.** It was explicitly skipped for want of the necessary numerical check and is not claimed here.
- 5) **Exact-zero covariance composition** holds; approximate leakage does not inherit it, and neither implies independence or a nonlinear guarantee.